1.3 Steering System Choices

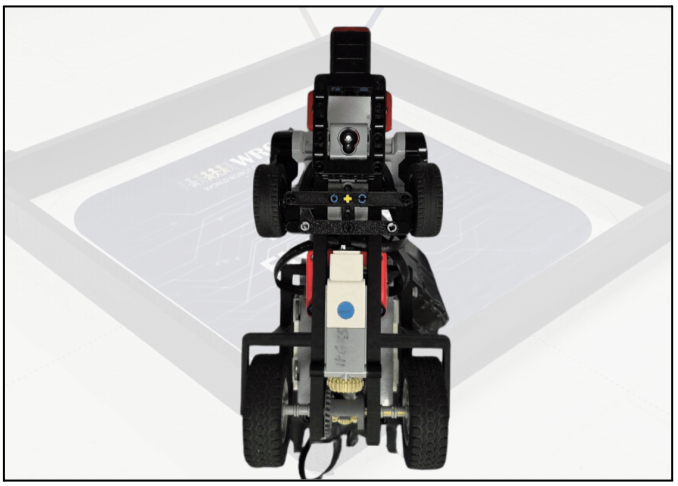
1.3.1 Steering Motor Selection

EV3 Medium Motor

6.64 N·cm
165 RPM
45503
Compact and lightweight; higher rotational speed

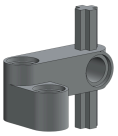
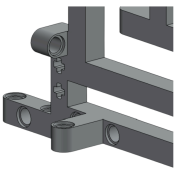
- We selected the LEGO EV3 Medium Servo Motor (45503) to operate our steering mechanism. It is vertically mounted to drive the steering linkage through LEGO Part 33299.
- The Medium Motor was chosen because of its compact size, sufficient torque for operating the steering mechanism, and responsive movement.

1.3.2 Steering Structure

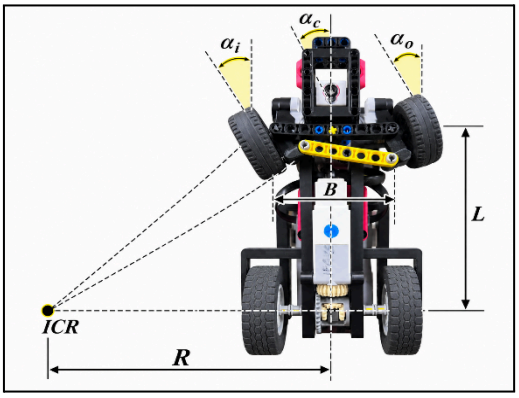


Our steering system combines LEGO Part 33299 with fully custom-designed 3D-printed components. Using LEGO axles and connectors, all components are assembled into a complete steering mechanism, ensuring proper alignment and coordinated movement of both front wheels.

Steering System Material

9 cm Main Section	
7 cm Perpendicular Section	
Lego 33299	
L-shaped Tie Rod x 2	
Integrated Chassis Mounting	
Rim & Tyre **Lego 56904 + Lego 30699**	

1.3.3 Ackermann Steering Geometry



- Our steering system follows the Ackermann steering principle, where the inner front wheel turns at a greater angle than the outer front wheel.
- As shown in our robot chassis, the steering linkages control both front wheels at different angles during cornering.
- The projected wheel axes meet at a common Instantaneous Centre of Rotation (ICR), allowing the wheels to follow a different turning radius.
- This reduces tyre scrubbing, minimizes unnecessary friction and improves the robot's turning efficiency and maneuverability.

Using our robot chassis as an example:

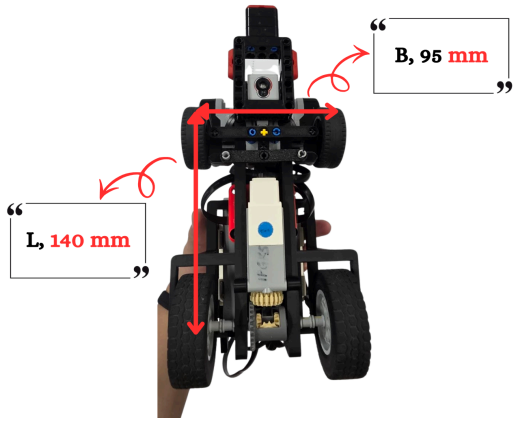
1. **Inner Wheel (α_i)** – During a left turn, the left front wheel is the inner wheel. It turns at a larger angle because it follows a smaller turning radius.

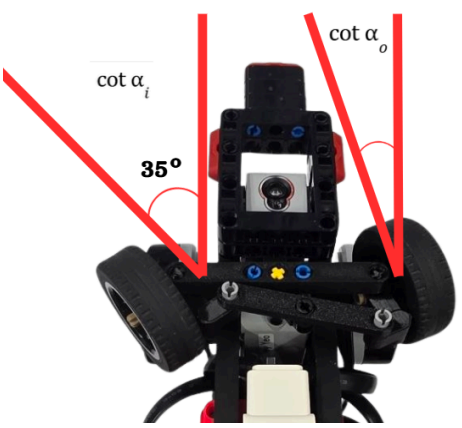
2. **Outer Wheel (α_o))** – The right front wheel is the outer wheel. It turns at a smaller angle because it follows a larger turning radius.

3. **Centre Steering Angle (α_c))** – Represents the reference steering angle at the centre of the front axle, between the inner and outer wheel angles.

4. **Instantaneous Centre of Rotation (ICR)** – The projected lines perpendicular to the direction of each wheel meet at a common point called the ICR. This allows all four wheels to follow their respective turning paths without unnecessary sideways slipping.

1.3.4 Ackermann Steering Testing and Calculation





$$\cot \alpha_o - \cot \alpha_i = \frac{B}{L}$$
$$\cot \alpha_o - \cot 35^\circ = \frac{95}{125}$$
$$\cot \alpha_o = 1.4281 + 0.76$$
$$\cot \alpha_o = 2.1881$$
$$\alpha_o = \tan^{-1}\left(\frac{1}{2.1881}\right)$$
$$\alpha_o = 24.57°$$

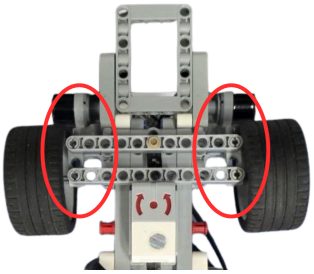
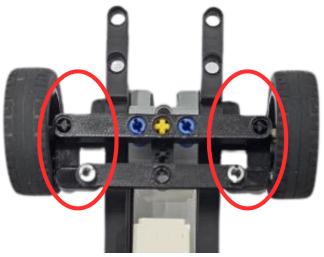
- During a left turn, the left front wheel turns at a larger angle of 35°, while the right front wheel turns at a smaller angle of 24.57°.
- This helps both front wheels follow their respective turning paths and provides smoother cornering.

1.3.5 Steering System Trade-off and Engineering Decisions

Advantages of Ackermann Steering System	Disadvantages of Ackermann Steering System
Better turning performance	Complex linkage design
Reduced tyre scrubbing	Additional components
Improve Manoeuvrability	Assembly effort

- The Ackermann steering system was selected because it provides better turning performance, reduces tyre scrubbing and improves manoeuvrability.
- Although it requires a more complex linkage design and additional assembly effort, its steering benefits outweigh these disadvantages.

Previous vs New Steering System:

Aspect	Previous Steering System	New Steering System
Picture		
Steering Structure	Two equal-length linkages	Ackermann steering linkage
Wheel Angle	Both front wheels turn at approximately the same angle	Inner wheel turns at a greater angle
Tyre Scrubbing	More likely during cornering	Reduced tyre scrubbing
Mechanical Complexity	Simple structure	More complex linkage design

- Despite the increased mechanical complexity and assembly effort, we selected the Ackermann steering system for its improved steering accuracy, reduced tyre scrubbing and smoother cornering.
- Overall, the performance benefits outweigh the additional design complexity, making it a better choice for our competition robot.